

MATH-334 Numerical Analysis

Numerical Differentiation-(Second Order)

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Topics Covered

① Taylor Series

② Numerical Differentiation

Forward Difference Appr. of 1^{st} Derivative

Backward Difference Appr. of 1^{st} Derivative

Centered Difference Appr. of 1^{st} Derivative

Second Order Differentiation

The Taylor Theorem and Series

Taylor theorem The Taylor theorem states that any smooth function can be approximated as a polynomial.

Taylor series The Taylor series provides a means to express this idea mathematically.

$$f(x) = f(x_0) + \frac{(x - x_0)}{1!} f'(x_0) + \frac{(x - x_0)^2}{2!} f''(x_0) + \cdots + \frac{(x - x_0)^n}{n!} f^{(n)}(x_0) + R_n$$

The remainder term R_n is of the order of h^{n+1} , meaning:

- The smaller the spacing, the smaller the error for a given number of terms.

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \cdots + \frac{f^{(n)}(x_i)h^n}{n!} + \cdots$$

Approximation of 1st order Derivative

Forward Difference Formula

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h}$$

Backward Difference Formula

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h}$$

Central Difference Formula

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h}$$

Step Size

h is called the **step size**, that is, the length of the interval over which the approximation is made.

For $f(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$

$f'(0.5) = -0.9125$. Accuracy in each case may be compared.

FD

$$\begin{aligned}
 f'(0.5) &= \frac{f_{i+1} - f_i}{h} \\
 &= \frac{0.2 - 0.925}{0.5} \\
 &= -1.45 \\
 |\epsilon_t| &= \\
 100 \cdot \frac{-0.9125 + 1.45}{-0.9125} \\
 |\epsilon_t| &= 58.9\%
 \end{aligned}$$

BD

$$\begin{aligned}
 f'(0.5) &= \frac{f_i - f_{i-1}}{h} \\
 &= \frac{0.925 - 1.2}{0.5} \\
 &= -0.55 \\
 |\epsilon_t| &= 39.7\%
 \end{aligned}$$

CD

$$\begin{aligned}
 f'(0.5) &= \frac{f_{i+1} - f_{i-1}}{2h} \\
 &= \frac{0.2 - 1.2}{1} \\
 &= -1 \\
 |\epsilon_t| &= 9.6\%
 \end{aligned}$$

Finite Difference Approximations of 2nd Derivatives.

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To do this, we write a forward Taylor series expansion for $f(x_{i+2})$ in terms of $f(x_i)$. We know

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \cdots, \quad \text{where } h = x_{i+1} - x_i.$$

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$$f(x_{i+2}) = f(x_i) + f'(x_i)(2h) + \frac{f''(x_i)(2h)^2}{2!} + \cdots + \frac{f^{(n)}(x_i)}{n!}(2h)^n + \cdots$$

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Multiplying second equation by 2 and subtracting from first.

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$$f(x_{i+2}) - 2f(x_{i+1}) = -f(x_i) + f''(x_i)h^2 + \dots$$

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$$f(x_{i+2}) - 2f(x_{i+1}) = -f(x_i) + f''(x_i)h^2 + \dots$$

Simplifying this we get

$$f''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2} + \mathbf{O}(h)$$

Approximation of 2nd order Derivative

Forward Difference Formula

$$f''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2} + \mathbf{O}(h)$$

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Similar manipulations can be employed to derive a backward version

Backward Difference Formula

$$f''(x_i) = \frac{f(x_i) - 2f(x_{i-1}) + f(x_{i-2}))}{h^2} + \mathbf{O}(h)$$

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Backward Difference Formula

$$f''(x_i) = \frac{f(x_i) - 2f(x_{i-1}) + f(x_{i-2}))}{h^2} + \mathbf{O}(h)$$

A centered version

Central Difference Formula

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2} + \mathbf{O}(h^2)$$

Central Difference Formula for 2nd Order Derivative

Taylor series expansion of $f(x_{i+1})$ and $f(x_{i-1})$ are given by

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \dots$$
$$f(x_{i-1}) = f(x_i) - f'(x_i)h + \frac{f''(x_i)h^2}{2!} - \dots$$

The addition

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The addition $f(x_{i+1}) + f(x_{i-1})$ gives

Central Difference Formula for 2nd Order Derivative

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$$\begin{aligned}f(x_{i+1}) &= f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \dots \\f(x_{i-1}) &= f(x_i) - f'(x_i)h + \frac{f''(x_i)h^2}{2!} - \dots\end{aligned}$$

The addition $f(x_{i+1}) + f(x_{i-1})$ gives

$$f(x_{i+1}) + f(x_{i-1}) = 2f(x_i) + f''(x_i)h^2 + \mathbf{O}(h^4)$$

Central Difference Formula for 2nd Order Derivative

Taylor series expansion of $f(x_{i+1})$ and $f(x_{i-1})$ are given by

$$f(x_{i+1}) = f(x_i) + f'(x_i)h + \frac{f''(x_i)h^2}{2!} + \dots$$

$$f(x_{i-1}) = f(x_i) - f'(x_i)h + \frac{f''(x_i)h^2}{2!} - \dots$$

The addition $f(x_{i+1}) + f(x_{i-1})$ gives

$$f(x_{i+1}) + f(x_{i-1}) = 2f(x_i) + f''(x_i)h^2 + \mathbf{O}(h^4)$$

or

$$f''(x_i) = \frac{f(x_{i+1}) + f(x_{i-1}) - 2f(x_i)}{h^2} + \mathbf{O}(h^2)$$

Example

The distance x of a runner from a fixed point is measured at intervals of half a second. The data obtained are

t	0.0	0.5	1.0	1.5	2.0
x	0.00	3.65	6.80	9.90	12.15 (in metres)

Use a central difference to approximate the runner's acceleration at $t = 1.5$ s.

Solution Our aim here is to approximate $x''(t)$.

$$f''(a) \approx \frac{f(a+h) - 2f(a) + f(a-h)}{h^2}$$

Using data with $t = 1.5$ s at its centre we obtain

$$\begin{aligned} x''(1.5) &\approx \frac{x(2.0) - 2x(1.5) + x(1.0)}{0.5^2} \\ &= -3.40 \text{ m s}^{-2}, \end{aligned}$$

from which we see that the runner is slowing down.

Use following to approximate the 2nd derivative of $f(x) = xe^x$ at $x = 2.0$ with $h = 0.1$.

x	1.8	1.9	2.0	2.1	2.2
$f(x)$	10.889365	12.703199	14.778112	17.148957	19.855030

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$$\begin{aligned} f''(2) &= \frac{1}{(0.1)^2} (f(2.1) - 2f(2) + f(1.9)) \\ &= 100 \cdot (12.703199 - 2(14.778112) + 17.148957) \\ &= 29.593200. \end{aligned}$$

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 \end{aligned}$$

Where as

$$f''(2) = 29.704275, \quad \text{for } h = 0.2.$$

Approximate the error in 2nd derivative of $f(x) = xe^x$ at $x = 2.0$ with $h = 0.1$ and $h = 0.2$.

x	1.8	1.9	2.0	2.1	2.2
$f(x)$	10.889365	12.703199	14.778112	17.148957	19.855030

$$f''(a) \approx \frac{f(a+h) - 2f(a) + f(a-h)}{h^2}$$

- 1 Repeat the exercise with Forward Difference formula
- 2 Repeat the exercise with Backward Difference formula

Using more terms of the Taylor series expansion, more accurate formulas can be obtained.

Forward Difference

First Derivative

Error

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_i)}{h}$$

$$O(h)$$

$$f'(x_i) = \frac{-f(x_{i+2}) + 4f(x_{i+1}) - 3f(x_i)}{2h}$$

$$O(h^2)$$

Second Derivative

$$f''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2}$$

$$O(h)$$

$$f''(x_i) = \frac{-f(x_{i+3}) + 4f(x_{i+2}) - 5f(x_{i+1}) + 2f(x_i)}{h^2}$$

$$O(h^2)$$

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Backward Difference

First Derivative

Error

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h}$$

$$O(h)$$

$$f'(x_i) = \frac{3f(x_i) - 4f(x_{i-1}) + f(x_{i-2})}{2h}$$

$$O(h^2)$$

Second Derivative

$$f''(x_i) = \frac{f(x_i) - 2f(x_{i-1}) + f(x_{i-2}))}{h^2}$$

$$O(h)$$

$$f''(x_i) = \frac{2f(x_i) - 5f(x_{i-1}) + 4f(x_{i-2}) - f(x_{i-3}))}{h^2}$$

$$O(h^2)$$

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Centered Difference

First Derivative

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1}))}{2h}$$

Error

$$O(h^2)$$

$$f'(x_i) = \frac{-f(x_{i+2}) + 8f(x_{i+1}) - 8f(x_{i-1}) + f(x_{i-2}))}{12h}$$

$$O(h^4)$$

Second Derivative

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1}))}{h^2}$$

$$O(h^2)$$

$$f''(x_i) = \frac{-f(x_{i+2}) + 16f(x_{i+1}) - 30f(x_i) + 16f(x_{i-1}) - f(x_{i-2}))}{12h^2}$$

$$O(h^4)$$